

WEEK 7 – STUDENT PERFORMANCE

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INTRODUCTION:

The Student Performance in Exams dataset was imported into Power BI for analysis. The data was cleaned and changed using Power Query. Student details and subject scores were checked and prepared for analysis. Calculated columns and other required changes were also made to understand student performance easily.

DATA IMPORTING:

The dataset was imported into Power BI and opened in Power Query. The rows and columns were checked to understand the data. The data types, missing values, errors, and duplicate records were also checked. This helped to identify any problems in the dataset before cleaning it.

StudentsPerformance.csv							
File Origin		Delimiter		Data Type Detection			
1252: Western European (Windows)		Comma		Based on first 200 rows			
gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
female	group B	bachelor's degree	standard	none	72	72	74
female	group C	some college	standard	completed	69	90	88
female	group B	master's degree	standard	none	90	95	93
male	group A	associate's degree	free/reduced	none	47	57	44
male	group C	some college	standard	none	76	78	75
female	group B	associate's degree	standard	none	71	83	78
female	group B	some college	standard	completed	88	95	92
male	group B	some college	free/reduced	none	40	43	39
male	group D	high school	free/reduced	completed	64	64	67
female	group B	high school	free/reduced	none	38	60	50
male	group C	associate's degree	standard	none	58	54	52
male	group D	associate's degree	standard	none	40	52	43
female	group B	high school	standard	none	65	81	73
male	group A	some college	standard	completed	78	72	70
female	group A	master's degree	standard	none	50	53	58
female	group C	some high school	standard	none	69	75	78
male	group C	high school	standard	none	88	89	86

Extract Table Using Examples

Load

Transform Data

Cancel

DATA CLEANING:

The dataset was cleaned by removing duplicate records and checking invalid values. The data types were corrected where needed. The categorical values were also standardized to keep the data clear and consistent.

123 Student ID	A ^B _C gender	A ^B _C race/ethnicity	A ^B _C parental level of education
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	Female	Group B	Bachelor'S Degree
2	Female	Group C	Some College
3	Female	Group B	Master'S Degree
4	Male	Group A	Associate'S Degree
5	Male	Group C	Some College
6	Female	Group B	Associate'S Degree
7	Female	Group B	Some College
8	Male	Group B	Some College
9	Male	Group D	High School
10	Female	Group B	High School
11	Male	Group C	Associate'S Degree

CALCULATED COLUMNS:

Calculated columns were created for Total Marks and Average Score. A Performance Grade was also created using conditional logic to classify student performance.

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula ⓘ

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula ⓘ

STUDENT PERFORMANCE GRADE AND RESULT:

A Performance Grade was created to classify students based on their scores. The grades show different levels of student performance. A Pass/Fail column was also created to identify whether the student passed or failed based on the selected pass criteria.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Grade

	Column Name	Operator	Value		Output
If	Average Score	is greater than or...	80	Then	A
Else If	Average Score	is greater than or...	60	Then	B
Else If	Average Score	is greater than or...	40	Then	C

... Add Clause

Else
Fail

OK Cancel

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Result

	Column Name	Operator	Value		Output
If	Average Score	is greater than or...	40	Then	Pass

... Add Clause

Else
Fail

OK Cancel

SUBJECT DATA TRANSFORMATION:

The Math, Reading, and Writing score columns were transformed using the Unpivot Columns option in Power Query. The new columns were renamed as Subject and Score to make the data easier to analyze. A separate duplicate table called Student Details was also created. This table contains only the student details and does not include the subject marks.

This helps to keep student information separate from the score data and makes the dataset easier to manage.

1 ² ₃ Student ID	A ^B _C Subject	1 ² ₃ Score
- Valid 100% - Error 0% - Empty 0%	- Valid 100% - Error 0% - Empty 0%	- Valid 100% - Error 0% - Empty 0%
1	Math Score	72
1	Reading Score	72
1	Writing Score	74
2	Math Score	69
2	Reading Score	90
2	Writing Score	88
3	Math Score	90
3	Reading Score	95
3	Writing Score	93
4	Math Score	47
4	Reading Score	57
4	Writing Score	44

1 ² ₃ Student ID	A ^B _C gender	A ^B _C race/ethnicity
- Valid 100% - Error 0% - Empty 0%	- Valid 100% - Error 0% - Empty 0%	- Valid 100% - Error 0% - Empty 0%
1	Female	Group B
2	Female	Group C
3	Female	Group B
4	Male	Group A
5	Male	Group C
6	Female	Group B
7	Female	Group B
8	Male	Group B
9	Male	Group D
10	Female	Group B
11	Male	Group C
12	Male	Group D

DAX MEASURES AND DATA MODEL:

The transformed data was prepared for the Power BI data model. DAX measures such as Average Math Score, Average Reading Score, Average Writing Score, Overall Average Score, and Overall Pass % were created.

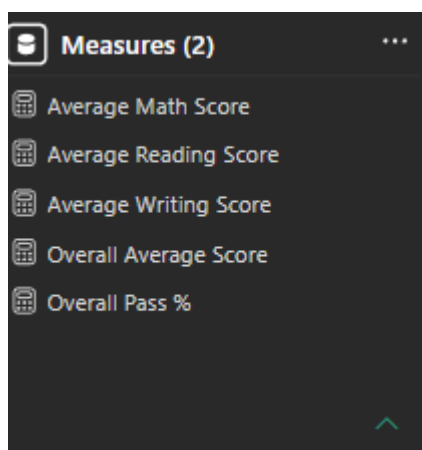
```
Average Math Score = AVERAGE('Student Performance'[math score])
```

```
Average Reading Score = AVERAGE('Student Performance'[reading score])
```

```
Average Writing Score = AVERAGE('Student Performance'[writing score])
```

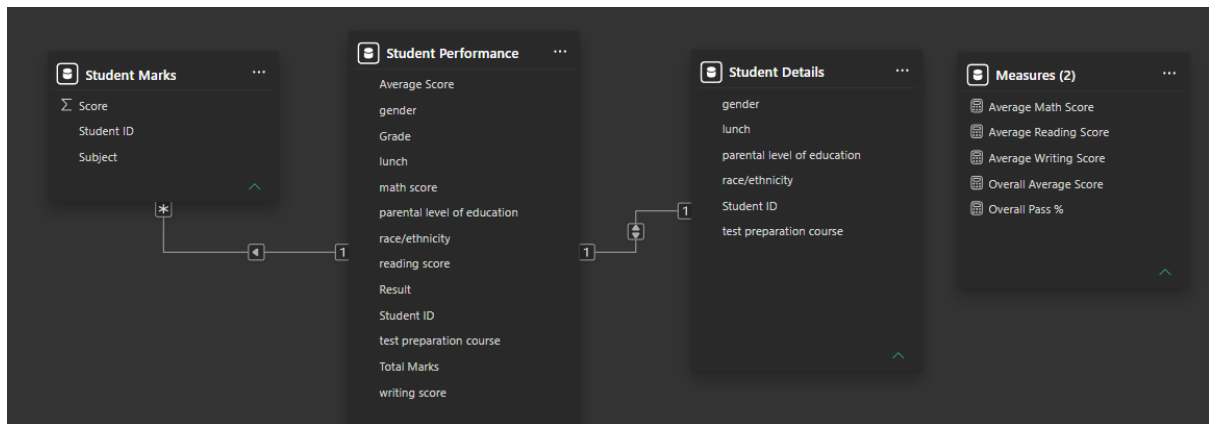
```
Overall Average Score = AVERAGE('Student Marks'[Score])
```

```
Overall Pass % = DIVIDE(CALCULATE(COUNT('Student Performance'[Result]), 'Student Performance'[Result]="Pass"),DISTINCTCOUNT('Student Performance'[Student ID]),0)
```



FINAL OUTPUT:

The final output is a cleaned and transformed Student Performance dataset. It contains calculated attributes such as Total Score, Average Score, Performance Grade, and Pass/Fail classification. The subject scores were also unpivoted, and basic DAX measures were created for further analysis.



CONCLUSION:

The Week 7 Student Performance dataset was successfully cleaned and transformed using Power Query and Power BI. Calculated columns, Performance Grades, and Pass/Fail classifications were created to understand student performance. The subject data was unpivoted for easier analysis, and DAX measures were added to calculate important performance values. The final dataset is now ready for visualization and further analysis.